PI7C9X3G606GP / PI7C9X3G808GP / PI7C9X3G816GP / PI7C9X3G1224GP PI7C9X3G1632GP
Diodes Gen3 PCIe® Packet Switch for FailOver Application

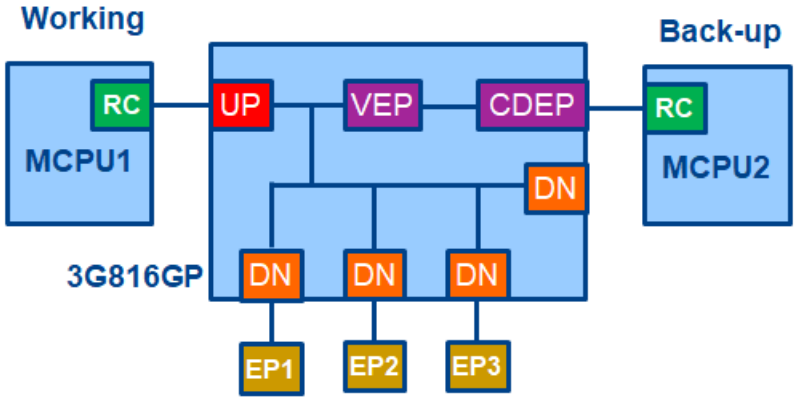
Introduction

The PCIe Packet Switch supports Cross-Domain End-Point (CDEP) function. The function, when working in conjunction with the software running on the PRIMARY HOST, enables data exchange among multiple PCIe-based hosts. This feature allows the system to implement fail-over functionalities, and the register setting flow is briefly described in the following sections with the suitable CDEP Mode Switch Model.

In an error resilient system, a back-up or secondary host is set up to take over the primary host in the event when primary host has failed and the failure is detected. The CDEP Mode Switch Model facilitates the data exchange across domains of the primary and secondary hosts, and swaps the roles of the hoses in the event of the primary host's failure.

In the event when the primary host is unable to send regular messages to the back-up host, the back-up host can swap its role to the primary host and its CDLEP port being an agent of upstream port to handle the traffic to/from downstream ports. The previous primary host's upstream port will be reset during the link-down process initiated by the previous primary host. The Switch has to ensure that the reset does not propagate to the entire upstream hierarchy in the previous primary host's, so that the new primary host can seamlessly manage the original End-Points. As a result, the impacts to the entire system are minimal during the fail-over transition.

The following illustration and settings use 3G816 for reference：



Fail-Over Operation:

- 3G816GP switch is configured as CDEP mode
- Once MCPU1 is down, the back-up MCPU2 is put in operation
  - 3G816GP switch is configured as Fan-Out Mode
  - At MCPU2 side, the up-port selection makes up-port swapped to face RC of MCPU2
  - At MCPU1 side, the link is down with 3G816GP
- After MCPU is repaired and resumes operation, the up-port selection will be restored to the original setting, which has RC of MCPU1 facing the upstream port. Also, the 3G816GP is configured back to CDEP mode

Initial Register Settings by EEPROM:

- Device Configuration 0 Register
  - 504h bit[6] = 1b (enable Chip CD mode, switch operates in cross-domain mode)
- Device Configuration 2 Register
  - 50Ch bit[17:16] = 11b (one CDVEP port and one CDLEP port)
- Device SRIS External Control Register
  - 514h bit[31] = 1b (Device SRIS external Control Enable)
- Fail-Over Control Register
  - 534h bit[2:1] = 11b (To “disable up link down, fire down port hot-reset event”, and to “enable up link down, keep up port enum data)
- CDEP DATA 1 Register
  - A08h bit[31] = 1b (the link between local host and CDLEP port is enabled. Once link is up, the configuration and memory commands will be sent to PRIMARY HOST via message queue or cross-domain translated transaction)

Software Setting Flow：

When mCPU1 at P0 side is dead, P4 will be changed to up port. Below are the procedures for software. Those commands can be done by CFG or I2C/SMBUS commands.

Fail-Over for upstream port from P0 to P4

1) Write LINK CONTROL register2 (98h bit[3:0]=1b) at port 0: set Port 0 target link speed to GEN1
2) Write LINK CONTROL register (78h bit[5]=1b) at port 0: retrain the link
3) Write LTSSM 0 register (390h bit[2]=1b) at port 0: force Port 0 LTSSM state stay in detect state
4) Write LINK CONTROL register2 (98h bit[3:0]=1b) at port 0x11: set Port 11 (CDEP, P4) target link speed to Gen1
5) Write LINK CONTROL register (78h bit[5]=1b) at port 0x11: retrain the link
6) Write LTSSM 0 register (390h bit[2]=1b) at port 0x11: force Port 11 (CDEP, P4) LTSSM state stay in detect state
7) Write CFG0 register (504h bit[6]=0b) at port 0: disable Chip CD Mode (chip operates in transparent mode, no cross domain mode)
8) Write CFG0 register (504h bit[2:0]=100b) at port 0: select Port 4 as upstream port[Note1]

9) Write CFG0 register (50Ch bit[17:16]=00b) at port 0: disable Switch CD Mode (no CDEP/CDBR ports configured in this switch)
10) Write PCIE switch class rev id (8h bit[23:16]=4h) at port 4: change sub class code from 0x80 (other bridge devices) to 0x04 (PCI to PCI bridge)
11) Write PCIE CAP register (68h bit[23:20]=5, bit[24]=0b) at port 4: to set port/device type/function to PCIe switch upstream port
12) Write PCIE CAP register (68h bit[23:20]=6, bit[24]=1b) at port 0: to set port/device type/function to PCIe switch downstream port
13) Write LINK CONTROL register2 (98h bit[3:0]=3b) at port 0: set Port 0 target link speed to GEN3
14) Write LTSSM 0 register (390h bit[2]=0b) at port 0: disable forcing Port 0 LTSSM state stay in detect state
15) Write LINK CONTROL register2 (98h bit[3:0]=3b) at port 4: set Port 4 target link speed to GEN3
16) Write LTSSM 0 register (390h bit[2]=0b) at port 4: disable forcing Port 4 LTSSM state stay in detect state
17) Change RC 18h：18h needs to re-setting as Port 4 becomes to upstream port
18) Write BUS Number register (18h) of Port 4 as Port 4 becomes to upstream port

When mCPU1 is back, P0 will change back as up port. Below are the procedures for software. Those commands can be done by CFG or I2C/SMBUS commands.

Fail-Over for upstream port from P4 to P0

1) Write LINK CONTROL register2 (98h bit[3:0]=1b) at port 0: set Port 0 target link speed to GEN1
2) Write LINK CONTROL register (78h bit[5]=1b) at port 0: retrain the link
3) Write LTSSM 0 register (390h bit[2]=1b) at port 0: force Port 0 LTSSM state stay in detect state
4) Write CFG0 register (504h bit[2:0]=000b) at port 0: select Port 0 as upstream port[Note1]
5) Write CFG0 register (504h bit[6]=1b) at port 0: enable Chip CD Mode (chip operates in cross domain mode, not in transparent mode)
6) Write CFG0 register (50Ch bit[17:16]=11b) at port 0: enable tile CD Mode with 2 CDEP ports (one CDVEP port and one CDLEP port as it master tile) (Port 4 change to Port 0x11)
7) Write LINK CONTROL register2 (98h bit[3:0]=1b) at port 0x11: set Port 11 (CDEP, P4) target link speed to Gen1
8) Write LINK CONTROL register (78h bit[5]=1b) at port 0x11: retrain the link
9) Write LTSSM 0 register (390h bit[2]=1b) at port 0x11: force Port 0x11 LTSSM state stay in detect state
10) Write PCIE switch class rev id (8h bit[23:16]=80h) at port 0x11: change sub class code from 0x04 (PCI to PCI bridge) to 0x80 (other bridge devices)
11) Write PCIE CAP register (68h bit[23:20]=5, bit[24]=0b) at port 0: to set port/device type/function to PCIe switch upstream port
12) Write PCIE CAP register (68h bit[23:20]=0, bit[24]=0b) at port 0x11: to set port/device type/function to PCIe switch downstream port
13) Write LINK CONTROL register2 (98h bit[3:0]=3b) at port 0x11: set Port 11 (CDEP, P4) target link speed to GEN3
14) Write LTSSM 0 register (390h bit[2]=0b) at port 0x11: disable forcing Port 11 (CDEP, P4) LTSSM state stay in detect state
15) Write LINK CONTROL register2 (98h bit[3:0]=3b) at port 0: set Port 0 target link speed to GEN3
16) Write LTSSM 0 register (390h bit[2]=0b) at port 0: Disable forcing Port 0 LTSSM state stay in detect state
17) Set RC 18h: 18h needs to re-setting as Port 0 becomes to upstream port
18) Write BUS Number register (18h) of Port 0 as Port 0 becomes to upstream port

Note1: Up Port Selection for PI7C9X3G1632GP (two tiles) or PI7C9X3G1224GP (1.5 tiles) Supplementary

For PI7C9X3G1632GP, if up port is not at T0_P0, except offset 504h, it also need set offset 558h for virtual up/dn mapping. Below is an example if up port is changed to T1_P0.

1) Write DEVICE CONFIGURATION 0 register, offset 504h
   bit[5:0]: up port selection, for Tile 0 and Tile 1, P0 (Tile0 = Tile1 setting)

2) Write SWITCH DOMAIN MODE CONTROL register, offset 558h
   bit[6:0]: enable virtual Up/Dn
   - If up is set to T0_P0
     T0_P0 (master), offset 558h, bit[0]=0, bit[1]=1
     T1_P0 (slave), offset 558h, bit[0]=1, bit[6:1]=0
   - If up is set to T1_P0
     T0_P0 (slave), offset 558h, bit[0]=1, bit[6:1]=0
     T1_P0 (master), offset 558h, bit[0]=0, bit[1]=1